

Seminar on Artificial Intelligence and Quantum Computing 2026

Thursday, 15 October 2026

9:00 Day 1 Registration

9:30 Welcome Remarks - *Oscar Montiel Ross*

9:40 **Quantum Image Segmentation for Water Body Detection in Grayscale Satellite Imagery.**

Sandra S. Rosales-Alvarado, Ulises Orozco-Rosas, Oscar Montiel, Sergio Ivvan Valdez-Peña.

Presenter: Sandra S. Rosales-Alvarado; **Email:** srosales@citedi.mx; **Institution:** CITEDI.

10:00 **Variational Quantum Eigensolver for Molecular Electronic Structure Calculations.**

Daniel Villegas Piña, Jesús Yaljá Montiel Pérez, Juan Humberto Sossa Azuela, Alejandro Monroy Azpeitia.

Presenter: Alejandro Monroy Azpeitia; **Email:** amonroya2025@cic.ipn.mx; **Institution:** CIC.

10:20 **Development of an Intelligent Platform for Wastewater Characterization and Treatment Recommendation Using Artificial Intelligence.**

Adrian Jinich, Alejandro Castañeda, Leonardo Calderón Castillo.

Presenter: Adrian Jinich; **Email:** l22211477@tectijuana.edu.mx; **Institution:** ITT.

10:40 **CUDA-Accelerated Simulation of Variational Quantum Circuits: A High-Performance Framework for Data-Conditioned VQE Benchmarking.**

Juan Pablo Acuña González, Moisés Sánchez Adame, Oscar Humberto Montiel Ross.

Presenter: Juan Pablo Acuña González; **Email:** jagonzalez@citedi.mx; **Institution:** CITEDI.

11:00 *Keynote Lecture*



The Bayesian operating characteristic curve for feature analysis applied to urban land cover change.

Dr. Sergio Ivvan Valdez Peña

Email: svaldez@centrogeo.edu.mx

Institution: Centro de Investigación en Ciencias de Información Geoespacial, A.C.

11:40 Coffee Break

12:00 *Keynote Lecture*



Higher Type Fuzzy Systems in Control and Robotic Applications.

Dr. Oscar Castillo

Email: ocastillo@tectijuana.mx

Institution: Instituto Tecnológico de Tijuana

12:40 **Experimental Analysis of Quantum Error Correction Integration in Hybrid Quantum-Classical Systems for Classification with Variational Quantum Circuits.**

Jordi Fabián González-Contreras, Jesús Yaljá Montiel-Pérez, Erik Zamora, Juan Humberto Sossa-Azuela.

Presenter: Jordi Fabián González-Contreras; **Email:** jgonzalezc2023@cic.ipn.mx; **Institution:** CIC.



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13:00 **Performance Analysis of YOLO Models for Deployment on Resource-Limited Edge Devices in Security Applications.**

Miguel Angel Lopez Ramirez, César Lepe Garcia, Ricardo David Valdez Amparo, Andres Carlos Valdez Amparo.

Presenter: Ricardo David Valdez Amparo; **Email:** mlopez@tectijuana.edu.mx; **Institution:** ITT.

13:20 **Physics-Informed Quantum Optimization with Python and QuTiP: From Tunneling Barriers to Ising/QUBO Energy Landscapes.**

Juan Pablo Acuña González, Moisés Sánchez Adame, Oscar Humberto Montiel Ross.

Presenter: Juan Pablo Acuña González; **Email:** acua6307@gmail.com; **Institution:** CITEDI.

13:40 **Artificial intelligence into optical three-dimensional imaging: advances and perspectives.**

Rigoberto Juarez-Salazar, Sofia Esquivel-Hernandez, Jose H. Godoy Gonzalez, Christian Gaxiola-Paez, Victor H. Díaz Ramirez, Carlos Robledo-Sanchez.

Presenter: Rigoberto Juarez-Salazar; **Email:** rjuarez@citedi.mx;

Institution: Benemérita Universidad Autónoma de Puebla (BUAP).

14:00 **Hierarchical Temporal Attention Networks for Anatomically-Informed Violence Detection in Surveillance Systems.**

Seyi Tope Ogunji, Wonder Bassey Paul, Moises Sanchez Adame, Oscar Humberto Montiel Ross.

Presenter: Seyi Tope Ogunji; **Email:** seyitope@citedi.mx; **Institution:** CITEDI.

14:20 **Gaussian Continuous-Variable Quantum Neural Networks for Medical Image Applications.**

Daniel Alejandro Lopez, Oscar Montiel, Miguel Lopez-Montiel, Oscar Castillo.

Presenter: Daniel Alejandro López Montiel; **Email:** dlopez@citedi.mx; **Institution:** CITEDI.

14:40 **Multi-Agent Roundabout Circulation and Lane Changing using Soft Actor-Critic Control and a Modified Van der Pol Reference.**

Jesus Alejandro Quiros Alcala, Ulises Orozco Rosas y Luis T. Aguilar.

Presenter: Jesus Alejandro Quiros Alcala; **Email:** jquiros@citedi.mx; **Institution:** CITEDI.

15:00 **Day 1 Concludes**



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